

INSTRUCTOR GUIDE

Urban Systems & Product Design

Course 02 — Facilitation Guide for Graduate Assistants and Faculty

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HOW TO USE THIS GUIDE

This guide supports graduate teaching assistants and faculty facilitating Course 02: Urban Systems & Product Design. Each session entry provides a Session Overview, Theoretical Frame, Discussion Arc with timing, numbered Discussion Questions for all students (with facilitation notes), Graduate Questions (G1–G2, teal-accented), Common Misconceptions, Cross-session Connections, and Facilitator Notes.

The course is designed for a dual-listed seminar (upper-division undergraduate + master's) drawing students from urban planning, public policy, MBA, and engineering programs. The graduated question structure makes the two levels separable if needed.

Patrick's practitioner voice in this course

Course 02 is grounded in Patrick Hoffman's experience building Social Bicycles (SoBi) — the product that became JUMP — and navigating the city-by-city regulatory and partnership landscape that urban mobility products require. Week 5 is structured around his @arcktip practitioner account and, when possible, his direct participation. Week 13 applies his go-to-market framework to student pitches. Throughout, his account of what cities actually want from products — and what they fear — is the practitioner counterpoint to the academic readings.

Pitch evaluation criteria — share at Week 1

Students should build toward these criteria from the first session:

- Urban systems insight — does the product demonstrate understanding of how cities work as complex systems?
- Equity commitment — is the commitment specific, measurable, and binding?
- Regulatory strategy — collaborative, realistic, and city-specific?
- City partner — named institutional partner with clear mutual benefit?
- Business model — honest about unit economics and externalized costs?
- Climate assessment — honest accounting of emissions impact and resilience contribution?

Week 1 · **Cities as Systems**

FORMAT	TIMING	LEVEL
Seminar	75 min	UG + Grad

SESSION OVERVIEW

This opening session frames cities not as static backdrops for products, but as dynamic, complex systems — networks of flows, feedback loops, and emergent behaviors that resist simple optimization. Students arrive with the intuition that urban problems have urban solutions; the goal is to complicate that intuition. Cities are not software to be debugged. They are adaptive systems shaped by millions of individual decisions, policy regimes, market forces, and physical constraints that interact in nonlinear ways. Products that enter this system must engage with it, not override it.

THEORETICAL FRAME

Jacobs is the anchor: her bottom-up, observational account of how cities actually work is the sharpest challenge to the top-down, optimization-oriented thinking that dominates tech product culture. Townsend provides the contemporary bridge — smart city technology as the latest iteration of the planner's impulse to rationalize the city. Batty's complexity science framing (for graduate students) provides the mathematical vocabulary for what Jacobs described intuitively.

SUGGESTED DISCUSSION ARC

0–10 min	Open with two satellite images: Manhattan's street grid and a single-use suburban office park. Ask: 'Which of these is a system? Which is a product?' Surface the distinction between designed artifacts and emergent complexity.
10–25 min	Brief lecture: cities as complex adaptive systems. Three key properties — emergence (the whole is more than the parts), nonlinearity (small inputs can have large effects), and path dependence (history matters). Jacobs identified all three without the vocabulary.
25–50 min	Seminar on Jacobs and Townsend. Questions 1–3. The 'is a city a system or a product?' question usually produces the most interesting disagreement.
50–65 min	Graduate section: Batty, Glaeser, Hall. Questions G1 and G2. The Glaeser question — what cities are fundamentally for — is foundational for the whole course.
65–75 min	Close: each student states one thing a product they have used has done to a city — good or bad. Collect on the board as the course's opening data set.

DISCUSSION QUESTIONS — ALL STUDENTS

- 1 **Jacobs argues that a city street is safe not because of police or design, but because of 'eyes on the street' — a complex web of mutual surveillance by residents who have reason to be there at different times. Can you design a product that produces eyes on the street? Or is that the wrong question?**

Goal: distinguish between design that enables complex behavior and design that tries to produce it directly. The latter almost always fails.

- 2 **Townsend describes smart city technology as a new form of the urbanist's impulse to understand and optimize cities. Jacobs spent her career opposing that impulse. Who's right?**

Don't resolve this — it's the course's animating tension. Students should leave with both arguments clearly in hand.

- 3 **If cities are complex systems that resist optimization, what is the appropriate role for a product built in a city? What can it do that top-down planning cannot? What can it not do?**

Forward-looking framing question. Students should return to their answer in Week 15.

GRADUATE / MASTER'S QUESTIONS

- G1 **Batty's 'new science of cities' applies network theory and complexity science to urban systems. Does that formalization help or hinder the kind of analysis Jacobs was doing? What is gained and what is lost when you mathematize urban complexity?**

- G2 **Glaeser argues that cities' productivity comes from the density of human interaction — the collision of ideas. If that's true, what is a city for? And what kinds of products reinforce that function vs. undermine it?**

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students from tech backgrounds often arrive with a strong optimization bias — cities have problems, products are solutions. Jacobs is the antidote. The goal is not to make students anti-product but to install healthy skepticism about simple interventions in complex systems.

Students from planning backgrounds sometimes over-celebrate Jacobs as anti-development. She was pro-density, pro-diversity, and pro-complexity — not anti-growth. Keep the analysis specific.

Graduate students may treat complexity science as the answer rather than a descriptive tool. It describes why cities resist optimization; it doesn't prescribe what to do about it.

CONNECTIONS TO OTHER SESSIONS

The complex systems framing established here is the analytical foundation for the whole course. Return to it explicitly every time a product is introduced: 'What does this product do to the system?' In Week 6 (platform urbanism), ask how platforms interact with urban feedback loops. In Week 14 (failure cases), ask which system properties the failed products ignored.

FACILITATOR NOTES

The satellite image opening works well even without prior preparation — Google Maps satellite view, any city. The contrast between a grid city (New York, Chicago, Portland) and a cul-de-sac suburb is immediately legible. For classes with strong international representation, use a city students know well — Lagos, Mumbai, and Tokyo all work beautifully as cases of emergent urban complexity.

Week 2 · **The Planning Tradition**

FORMAT Seminar	TIMING 75 min	LEVEL UG + Grad
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SESSION OVERVIEW

Before students can critique planning, they need to understand it — what planners actually do, why zoning was invented, what problems master plans were designed to solve, and why those solutions so often produced new problems. Scott's 'Seeing Like a State' provides the theoretical framework: high-modernist planning fails because it prioritizes legibility and control over the local knowledge that makes cities work. Bertaud offers the corrective: markets and planning are not opposites but complements, and planners who ignore price signals consistently produce bad outcomes.

THEORETICAL FRAME

The session deliberately presents the planning tradition charitably before critiquing it. The 1909 zoning codes were responses to real problems — factory fumes in residential blocks, overcrowding, sanitation failures. The critique is not that planners were wrong to plan, but that the tools they chose — comprehensive plans, use separation, top-down control — often made things worse. Bertaud's market-and-design synthesis is the productive middle ground.

SUGGESTED DISCUSSION ARC

0–10 min	Show a 1916 New York City zoning map and a 2024 one. Ask: 'What was the problem these maps were solving? Is that problem still the problem?' Surface: zoning's origins in industrial nuisance law vs. its current role in exclusion.
10–20 min	Brief lecture: the five tools of the planning tradition — comprehensive plans, use zoning, density zoning, design review, and environmental review. Each was invented to solve a problem; each created new ones.
20–50 min	Seminar on Jacobs, Scott, and Bertaud. Questions 1–3. The Jacobs vs. Bertaud tension (Question 3) is where the richest intellectual disagreement lives.
50–65 min	Graduate section: Hall, Hirt, Flyvbjerg. Questions G1 and G2.
65–75 min	Policy exercise: hand students the zoning code for one block in the course city. What can be built there? What cannot? Is that the right answer in 2026?

DISCUSSION QUESTIONS — ALL STUDENTS

- 1 | **Scott argues that high-modernist planning fails because it privileges 'legibility' —**

making places readable and controllable from above — over the local, practical knowledge ('metis') that makes things actually work. Find a contemporary product that makes a similar mistake.

Good cases: ride-hailing surge pricing that ignores local knowledge of when and where people need to go; algorithmic restaurant recommendations with no understanding of neighborhood context.

2 Zoning was invented to separate factories from homes. Most American cities now use it to separate everything from everything else — including affordable apartments from expensive ones. How did a tool designed to prevent harm become a tool that produces it?

The evolution from nuisance law to exclusionary zoning is the key story. Push students to trace the mechanism, not just observe the outcome.

3 Bertaud says markets and planning are complements, not opposites — planners should use price signals rather than fight them. Jacobs would agree on the emergence point but distrust markets as much as planners. Is there a synthesis?

There is no clean answer. The goal is for students to hold both positions clearly and begin developing their own.

GRADUATE / MASTER'S QUESTIONS

G1 Hirt argues that American zoning's separation of uses is unique globally — other rich democracies zone for density and character, not primarily use. What does that comparison tell us about American urbanism? And about which planning models are exportable?

G2 Flyvbjerg argues that planning decisions are almost never made by rational analysis — they are made by power, and rational analysis is deployed post hoc to justify the decision. If that's true, what is the point of evidence-based planning? What role should analysts play?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often conflate 'planning failed' with 'we should have no planning.' Bertaud's position — that markets and planning complement each other — is the productive synthesis. Zoning reform, not zoning abolition, is the policy implication.

Jacobs's critique of planning is sometimes read as Jacobs being anti-density. She was ferociously pro-density — her critique was of homogeneous superblock development, not of height or closeness per se.

Scott's 'Seeing Like a State' is sometimes read as libertarian. It is not — Scott is a left-anarchist. His critique is of bureaucratic knowledge, not of government per se.

CONNECTIONS TO OTHER SESSIONS

The planning tradition session is the historical baseline. Every subsequent session asks: what does this product or technology do to the planning regime? In Week 6 (platform urbanism), platforms operate in the gaps that zoning creates. In Week 9 (regulation as design), students see zoning codes as product design constraints. In Week 14 (failure cases), Sidewalk Toronto is a high-modernist planning failure dressed in tech-company clothes.

FACILITATOR NOTES

The zoning code exercise works best with a real, locally relevant example. Pull the actual zoning ordinance for a block students walk past on the way to class. The gap between what is legally allowed and what exists is always instructive — often the existing building would be illegal to build today.

Week 3 · Data & the City

FORMAT
Seminar

TIMING
75 min

LEVEL
UG + Grad

SESSION OVERVIEW

Data-driven urbanism has become the dominant mode of urban tech: dashboards monitoring traffic flows and 311 calls, algorithms predicting where crime will occur, models optimizing bus schedules. This session interrogates that mode — not to reject quantitative analysis of cities, but to specify what it can and cannot see. Mattern's 'A City Is Not a Computer' is the sharpest short statement of the critique; O'Neil and Eubanks ground it in empirical evidence of harm.

THEORETICAL FRAME

The session's central distinction is between data as tool and data as epistemology. Used as a tool, quantitative analysis can reveal patterns invisible to individual observation. Adopted as an epistemology — the belief that what cannot be measured doesn't matter — it systematically erases the most important features of urban life. The goal is for students to be sophisticated users of data, not opponents of it.

SUGGESTED DISCUSSION ARC

0–10 min

Show a crime heat map of any American city. Ask: 'What does this map show? What does it hide? Who made it, and for whom?' Surface: the map shows police activity, not crime; it was made with police data; it will be used to direct more police to already over-policed areas.

10–20 min

Brief lecture: three modes of urban data — descriptive (what happened), predictive (what will happen), prescriptive (what should happen). Each mode raises different accountability questions.

20–50 min

Seminar on Mattern, Green, and O'Neil. Questions 1–3. Question 2 (measurement displacement) is the conceptual core.

50–65 min

Graduate section: Kitchen, Eubanks, Loukissas. Questions G1 and G2.

65–75 min

Product exercise: each student names one metric their hypothetical urban product would track. Then: 'What does optimizing for that metric make worse?' Exchange papers and critique.

DISCUSSION QUESTIONS — ALL STUDENTS

- 1 **Mattern argues that calling a city 'smart' usually means making it legible to computational systems, not making it more responsive to residents. What would a city that was genuinely smarter — wiser, more responsive, more just — look like? What would you measure?**

Students often jump to better metrics. Push harder: is the problem the metrics, or the measurement orientation itself?

- 2 **O'Neil describes 'measurement displacement' — optimizing for a proxy metric instead of the thing you actually care about. Find an urban example. What is the thing you actually care about, and what proxy has replaced it?**

Classic examples: police metrics (arrests vs. safety), school ratings (test scores vs. learning), transit metrics (on-time performance vs. access). Students often have strong personal examples.

- 3 **If a city government wants to use data to better serve residents, what principles should govern how that data is collected, analyzed, and acted upon? Draft three principles.**

Normative question — no right answer, but students should be able to justify each principle.

GRADUATE / MASTER'S QUESTIONS

- G1 **Loukissas argues that 'all data are local' — data always reflects the conditions of its collection and cannot be meaningfully decontextualized. If that's true, how should urban product teams think about data portability? About scaling a product from one city to another?**

- G2 **Eubanks documents how automated decision systems harm poor and working-class communities specifically — not through explicit discrimination but through proxy variables that correlate with class. How should a product team audit their system for this kind of harm before deploying in a city?**

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often read the data critique as 'data is bad.' The argument is more specific: data is a representation, not the thing itself, and optimizing for the representation can degrade the thing. Data literacy, not data rejection, is the lesson.

The crime heat map example provokes strong political reactions. Keep the analysis methodological first: what does the map actually measure? Let the politics follow once the methodology is clear.

Graduate students sometimes treat Mattern as definitively settling the smart city debate. She is making a corrective argument, not a comprehensive one. Townsend's defense of civic data uses is the productive counterpoint.

CONNECTIONS TO OTHER SESSIONS

The data critique recurs throughout the course. In Week 5 (micromobility), GPS data from dockless bikes raises exactly these questions — who owns it, who can see it, what does it actually measure? In Week 8 (equity), algorithmic decision-making is the mechanism of discrimination. In Week 10 (civic tech), participatory budgeting is a response to the data-as-epistemology problem.

FACILITATOR NOTES

The product metrics exercise is highly effective. Give students 2 minutes to write their metric, then swap papers. The act of critiquing someone else's metric is more educational than critiquing your own. Save the paired critiques — use them in Week 12 pitch workshop to ask 'how has your thinking about metrics evolved?'

Week 4 · Mobility Systems I

FORMAT

Seminar

TIMING

75 min

LEVEL

UG + Grad

SESSION OVERVIEW

Transit is the connective tissue of cities: it determines who can reach which jobs, schools, hospitals, and neighborhoods. This session establishes the technical and political foundations of transit analysis before students encounter micromobility (Week 5) and platform mobility (Week 6). Walker's 'Human Transit' is the most useful single text for non-planners: it translates complex transit concepts into product-relevant terms.

THEORETICAL FRAME

The central technical concept is induced demand: adding road capacity generates new traffic, not less congestion. Students from product backgrounds often assume that adding service solves access problems; Walker's frequency vs. coverage trade-off shows that it is rarely that simple. The political economy dimension (why are American transit systems so poor?) connects directly to the Week 2 planning critique.

SUGGESTED DISCUSSION ARC

0–10 min

Ask: 'How did you get here today? If you could not use that mode, what would you do?' Map the class's mobility dependence. Surface: transit quality varies enormously by neighborhood, and that variation is not accidental.

10–20 min

Brief lecture: Walker's three fundamental transit trade-offs — frequency vs. coverage, speed vs. access, and cost vs. quality. Every transit product decision sits somewhere on these trade-offs. There is no free lunch.

20–50 min

Seminar on Jacobs (Ch. 18), Walker, and Vanderbilt. Questions 1–3. Question 2 (induced demand) is the one that most surprises product students.

50–65 min

Graduate section: Downs, Cervero, Levinson. Questions G1 and G2.

65–75 min

Map exercise: using the course city's transit map, identify three neighborhoods with high transit access and three with low access. Are the patterns correlated with income? With race? What product opportunity or policy gap does that reveal?

DISCUSSION QUESTIONS — ALL STUDENTS

1

Walker argues that frequency is the most important single factor in transit usefulness — a bus that comes every 5 minutes is categorically different from one that comes every 30

minutes. Why is frequency so hard to provide? Who decides?

The political economy answer: frequency costs money; coverage is more politically popular because it serves more constituencies, even if it means less frequency everywhere.

2**Induced demand means that adding road capacity generates new traffic — within years, congestion returns to its prior level. If that's true for roads, does the same logic apply to transit? To bike lanes? To dockless bikes?**

This is genuinely contested. The evidence for induced demand in transit is weaker than for roads. Let students work through what would have to be true for it to apply and what would have to be true for it not to.

3**What would a truly good transit system look like for the city where this course is taught? What would it require — technically, politically, financially? What product or technology could contribute, and what couldn't?**

Applied question grounding the theoretical discussion. Students should leave with a concrete sense of the gap between what transit could be and what it is.

GRADUATE / MASTER'S QUESTIONS**G1**

Cervero's 'Transit Metropolis' compares cities globally and argues that the most successful transit systems are built around transit-oriented development — density at stations that generates ridership. What does that finding imply for product design? Can a mobility app substitute for land use?

G2

Levinson and Krizek argue that digital mobility — autonomous vehicles, on-demand transit, mobility as a service — will fundamentally restructure the geometry of access. Does their argument depend on solving the equity problem first, or does it sidestep it?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often conflate transit quality with technology. Better buses don't require autonomous vehicles; they require frequency, reliability, and coverage. Technology is secondary to operations and funding.

Induced demand is often invoked as a trump card against all road investment. The empirical picture is more nuanced — the strength of the effect varies by context. Teach students to evaluate the evidence, not apply the heuristic automatically.

Graduate students from economics backgrounds sometimes treat transit as a market failure problem. Walker's political economy argument is that transit markets fail structurally, not just because of unpriced externalities.

CONNECTIONS TO OTHER SESSIONS

Mobility Systems I is the theoretical foundation for Week 5 (micromobility as a mobility product) and Week 13 (go-to-market — which cities have transit ecosystems where micromobility complements rather than replaces transit). The frequency/coverage trade-off returns in Week 12 (product pitch — students should specify where their product sits on this trade-off).

FACILITATOR NOTES

The transit map exercise is among the most powerful in the course. In cities with strong public transit (New York, Chicago, Boston, San Francisco), the equity patterns are visible and immediate. In car-dependent cities, the map reveals a different but equally important pattern: vast areas with no transit at all. Either way, the data is in the room.

Week 5 · Mobility Systems II — Micromobility

FORMAT

Seminar + Guest Practitioner

TIMING

90 min

LEVEL

UG + Grad

SESSION OVERVIEW

This is the course's primary practitioner session. Patrick Hoffman's account of building Social Bicycles (SoBi) — the product that became JUMP — is a first-hand record of how a mobility product engages with urban systems, regulatory environments, equity considerations, and business model pressures simultaneously. If Patrick can join, this session runs as a structured conversation. If not, the @arcktip essay is the primary text and the discussion structure still works.

THEORETICAL FRAME

The SoBi/JUMP arc is a complete product lifecycle in miniature: a small team with a design insight (dockless bikes using GPS for flexible docking) that navigated city by city, regulatory challenge by regulatory challenge, until acquisition by Uber — and what happened after. It demonstrates every concept from Weeks 1–4 in practice: urban complexity, regulatory planning tradition, data questions, and mobility trade-offs all appear in a single product story.

SUGGESTED DISCUSSION ARC

0–10 min

If Patrick is joining: introduce him and the @arcktip essay. If not: brief summary of the SoBi/JUMP arc. Either way, ask: 'What problem was Social Bicycles solving? For whom? At what point did it become a city problem and not just a technology problem?'

10–40 min

If Patrick is guest: structured conversation using the questions below. If not: seminar on Hoffman essay, Fishman, and NACTO. Questions 1–3 work in either format.

40–65 min

Graduate section: Sheller, McNeil, SFMTA pilot report as primary document. Questions G1 and G2. The SFMTA report is a primary policy document — treat it accordingly.

65–80 min

Design exercise: 'You are building the next dockless mobility product for [course city]. You have \$5M, 18 months, and one city to start in. What is your equity commitment, your regulatory strategy, and your first-year success metric? Walk me through it.'

80–90 min

Debrief the design exercise. What trade-offs did teams make? What did they prioritize? What did they sacrifice?

DISCUSSION QUESTIONS — ALL STUDENTS

1

The dockless innovation seems simple: bikes without stations, parked anywhere via GPS. But the regulatory and spatial implications were enormous. What assumptions

about urban space and city governance had to change for dockless bikes to become legal?

The question 'who owns the sidewalk?' was not previously asked in most American cities. Dockless bikes forced an answer. That forced answer is itself a product outcome.

2 Bikeshare systems consistently underserve low-income neighborhoods. Is that a technology problem, a business model problem, or a governance problem? What evidence would help you distinguish between these three explanations?

Evidence question: if it's technology, the fix is technical. If business model, the fix requires subsidy or different pricing. If governance, the fix requires regulatory change. Students should specify what evidence they would need for each hypothesis.

3 When JUMP was acquired by Uber, what happened to the product's mission? What does that acquisition tell us about the relationship between product values and capital structure?

Acquisition as a product event: the values of a product are partly determined by who owns it. What obligations do founders have to cities when they sell?

GRADUATE / MASTER'S QUESTIONS

G1 Sheller's mobility justice framework asks not just 'who can move?' but 'who has the right to mobility, and who decides?' Apply that framework to the JUMP product at the moment of Uber acquisition. What rights were implicated? Who held them?

G2 The SFMTA scooter pilot report is a government document. What does it measure, and what does that measurement reveal about what the city values? What does it not measure, and why might that be a political choice?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students sometimes read the equity critique of bikeshare as an argument against bikeshare. The argument is more demanding: equity is a condition of legitimacy for any mobility product operating in public space. That raises the bar, not the veto.

The 'Uber acquisition destroyed the mission' narrative is too simple. Uber invested in and scaled JUMP significantly before strategic pivots. The story is more complicated and more instructive for that complication.

Graduate students may be skeptical of practitioner accounts as theoretically unrigorous. Push back: Hoffman's account of navigating San Francisco's permitting process in real time is empirical data that no academic paper contains.

CONNECTIONS TO OTHER SESSIONS

This session is the course's applied fulcrum. The urban systems framing (Week 1), the planning critique (Week 2), the data questions (Week 3), and the mobility foundations (Week 4) all crystallize in the dockless bike story. Connect forward to Week 6 (platform urbanism — Uber is a platform, and JUMP's fate within Uber is a platform story) and Week 13 (go-to-market — the city-by-city regulatory strategy SoBi used is the template students should adapt for their pitches).

FACILITATOR NOTES

Guest practitioner format for Patrick: prepare him with 3–4 specific questions in advance. The most productive opening questions: (1) 'What was the moment the product became a city problem and not just a technology problem?' (2) 'What did cities get systematically wrong in their regulatory response to dockless bikes?' (3) 'If you were building JUMP today — after Uber, after the pandemic, with the current regulatory landscape — what would you do differently?' Allow students to ask follow-ups after each exchange.

Week 6 · Platform Urbanism

FORMAT
Seminar

TIMING
75 min

LEVEL
UG + Grad

SESSION OVERVIEW

Platform capitalism applies a specific business logic to urban space: extract data from interactions between users, benefit from network effects, and externalize costs onto infrastructure, workers, and neighborhoods. When Airbnb, Uber, and DoorDash apply this logic to housing, mobility, and food, the spatial consequences are profound and often invisible until they accumulate. This session makes those consequences legible and connects them to the product design decisions that produced them.

THEORETICAL FRAME

Wachsmuth and Weisler's 'Airbnb and the Rent Gap' is the session's empirical anchor: a careful spatial analysis of how Airbnb restructures housing markets neighborhood by neighborhood. This is the kind of rigorous, spatial thinking the course aims to produce. Srnicek provides the political economy framework; Rosenblat adds the labor and space dimensions of platform work.

SUGGESTED DISCUSSION ARC

0–10 min

Ask: 'Has anyone in this room stayed in an Airbnb, taken an Uber, or had food delivered in the last week?' Nearly all hands go up. 'You've participated in platform urbanism. What did you do to the city?' Let students respond — this usually produces surprised silence, then conversation.

10–20 min

Brief lecture: the three mechanisms of platform urbanism. (1) Data extraction: every transaction generates data used as competitive advantage. (2) Network effects: the more users, the more valuable the platform — until it dominates. (3) Externalization: platforms internalize profits and externalize costs onto streets, workers, and housing markets.

20–50 min

Seminar on Srnicek, Wachsmuth, and Rosenblat. Questions 1–3. Question 1 (rent gap mechanism) requires close reading and careful logic.

50–65 min

Graduate section: Zuboff, van Doorn, Sadowski. Questions G1 and G2.

65–75 min

Policy exercise: 'You are a city council member in a city with a 3% residential vacancy rate and a rapidly growing Airbnb market. What do you regulate, and how? What tools do you have?' Groups of 3, then share back.

DISCUSSION QUESTIONS — ALL STUDENTS

1 Wachsmuth and Weisler argue that Airbnb doesn't respond to the rent gap — it creates it. Walk me through that causal chain. How does a platform application shift the economic logic of an entire housing market?

Steps: Airbnb raises the revenue potential of short-term rental → landlords convert units → long-term supply decreases → rents rise → more landlords convert. Each step is a product outcome. The product team made decisions that enabled this chain. Were they responsible for it?

2 Uber and Lyft added enormous vehicle miles traveled to urban cores, increasing congestion in cities they claimed to be reducing congestion in. How is that possible? Who absorbed the cost?

Externalization of congestion costs onto streets and transit systems. The platform internalizes revenue from each ride and externalizes the congestion externality onto every road user.

3 Srnicek defines platforms by three features: they mediate between groups, they benefit from network effects, and they extract data. Does your proposed urban product have these features? Is that a problem?

Students designing products for cities often discover they are building platforms without having called them that. The question forces clarity.

GRADUATE / MASTER'S QUESTIONS

G1 Zuboff's surveillance capitalism describes human experience as the raw material for behavioral prediction products. Apply that framework to Uber. What is the raw material? What is the prediction product? Who buys it?

G2 Van Doorn argues that platform labor is specifically gendered and racialized — not by accident but by design. What spatial implications does that claim have? Where do racialized platform workers work, wait, and move through the city? Who sees them?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often frame platform critique as anti-market or anti-technology. Srnicek's argument is specifically about the platform business model, not about markets or digital technology in general. Keep the analysis structural.

The Airbnb rent gap argument applies differently in different markets — in tight housing markets the effect is large and well-documented; in loose markets it is weaker. Students should apply the framework to their specific city.

Graduate students sometimes treat Zuboff as totalizing or unfalsifiable. Acknowledge the critique while taking seriously the empirical documentation. Ask: what evidence would falsify her claim that surveillance capitalism is structurally extractive?

CONNECTIONS TO OTHER SESSIONS

Platform urbanism connects in every direction: back to Week 3 (data extraction as the mechanism) and forward to Week 8 (equity — who are the hosts and drivers?), Week 9 (regulation — what tools do cities have?), and Week 14 (failure cases — when platform urbanism fails at scale). The 'who owns the curb?' question from Weeks 4–5 becomes 'who owns the data from the curb?' in this session.

FACILITATOR NOTES

The policy exercise is most productive when students have looked up their city's short-term rental regulations before class — assign this as a 15-minute pre-class task. The contrast between cities with strong regulations (New York, Barcelona, Amsterdam) and those without is immediately instructive and produces good debate.

Week 7 · Field Session

FORMAT

Field observation

TIMING

3 hours

LEVEL

UG + Grad

SESSION OVERVIEW

The field session is the course's only non-seminar session and one of its most formative. Students move through the city with a structured observation brief, applying the analytical tools from Weeks 1–6 to the physical environment in real time. The brief is intentionally open-ended but analytically demanding: students must map mobility flows, identify friction points, locate equity gaps, and produce a 500-word field report before the next session. Whyte's Social Life of Small Urban Spaces is the methodological model.

THEORETICAL FRAME

Whyte developed his street observation method by watching — with cameras and notebooks — what people actually did in urban spaces, not what planners designed them to do. His findings consistently surprised planners who had designed those spaces. The field session asks students to practice the same discipline: systematic, patient observation before analysis.

SUGGESTED DISCUSSION ARC

0–15 min

Depart briefing: review the observation framework. Divide into teams of 2–3. Each team receives a specific block or intersection. Assign: (1) count pedestrian flows by direction, (2) map where people pause and why, (3) identify mobility friction points, (4) note equity gaps in who is using the space and how.

15–120 min

Field observation. Instructor and TAs circulate between teams. Students should observe in silence for at least 20 minutes before discussing what they see.

120–150 min

Teams reconvene at a central location. Each team shares three observations — one surprising, one confirming a course concept, one raising a question. Instructor synthesizes across teams.

150–180 min

Debrief and field report assignment. Reports due before next session. 500 words: what did you see, what course concept does it connect to, and what product opportunity or policy gap does it reveal?

DISCUSSION QUESTIONS — ALL STUDENTS

1

Whyte found that the best-used urban spaces were not those with the most amenities but those with the most 'triangulation' — elements that provoked strangers to interact. Did you observe triangulation in your field location? What produced it or prevented it?

Good examples of triangulation: street performers, food carts, interesting objects, water features. Good examples of anti-triangulation: blank walls, wide streets, surveillance cameras.

- 2** **Lynch's 'Image of the City' describes five elements that make cities legible: paths, edges, districts, nodes, and landmarks. Which of these were present in your observation site? Which were absent? How did their presence or absence affect how people moved?**

Applies primarily in the debrief. Ask students to categorize what they saw before applying the framework.

- 3** **You observed a real urban space for two to three hours. What surprised you most? What did the readings predict that you did not see? What did you see that the readings did not predict?**

The gap between theory and observation is the most valuable pedagogical moment. Protect it — don't resolve it too quickly.

GRADUATE / MASTER'S QUESTIONS

- G1** Appleyard's 'Livable Streets' showed that streets with higher traffic volumes had radically fewer social interactions among residents. Did you observe the Appleyard effect? What data would you need to test his claim rigorously in your observation site?

- G2** Your field report should connect observation to product opportunity or policy gap. As you write, ask: what would a product or policy intervention need to know about this space to serve it well? What would it need to not assume?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often want to immediately analyze rather than observe. Enforce the 20-minute observation period. The most valuable observations come after the obvious things have been noticed and students start seeing what is not there.

Urban observation can feel voyeuristic. Brief students on ethical field observation: no photography of individuals without consent, no entering private spaces, no recording conversations. The goal is spatial and behavioral pattern, not individual data.

Graduate students sometimes produce overly theorized field reports that are thin on observation. The model is Whyte: specific counts, specific times, specific behaviors. Theory comes second.

CONNECTIONS TO OTHER SESSIONS

The field session sits at the course's midpoint and grounds everything that follows. In Week 8 (equity), students return to their field site data and ask who was not there. In Week 12 (product pitch), they use their field observations as user research. In Week 15 (final pitch), the strongest pitches will cite specific field evidence from this session.

FACILITATOR NOTES

Site selection matters enormously. Choose a location with high pedestrian activity, visible mobility diversity (bikes, scooters, pedestrians, transit), and some visible equity dynamic. A busy commercial corridor near a transit stop in a mixed-income neighborhood is ideal. Ask students to bring notebooks, not phones — the act of writing slows down perception in productive ways.

Week 8 · Equity & Access

FORMAT

Seminar

TIMING

75 min

LEVEL

UG + Grad

SESSION OVERVIEW

Every urban product operates in a city that is already profoundly unequal. Transportation access, housing quality, digital connectivity, and exposure to environmental risk are all distributed along the lines of race, class, gender, and disability — not by accident but through deliberate policy choices accumulated over decades. This session foregrounds those inequalities as the context within which every urban product operates. The question is not whether a product will interact with urban inequality; it is how.

THEORETICAL FRAME

Rothstein's 'Color of Law' establishes that American residential segregation was government-produced — the result of FHA redlining, exclusionary zoning, and urban renewal, not natural market outcomes. That history is the spatial context within which every urban product is deployed. Sheller's mobility justice framework extends this to movement: the right to move is not equally distributed, and products that ignore that inequality tend to deepen it.

SUGGESTED DISCUSSION ARC

0–10 min

Return to the field session data. Ask: 'Who was in your observation site? Who was not? What modes of mobility did you see? What modes were invisible?' Use the observation data as the opening case.

10–20 min

Brief lecture: three mechanisms by which urban products can deepen inequality — by design (inaccessible to low-income users), by deployment (only in high-income areas), or by displacement (raises neighborhood value, pricing out existing residents).

20–50 min

Seminar on Rothstein, Sheller, and Klinenberg. Questions 1–3. Question 2 (design vs. deployment vs. displacement) is the most analytically productive.

50–65 min

Graduate section: Eubanks, Grengs, Ramiller. Questions G1 and G2.

65–75 min

Product design exercise: take a product from any previous session. Redesign it with equity as the primary design constraint. What changes? What stays the same? What trade-offs appear?

DISCUSSION QUESTIONS — ALL STUDENTS

- 1** Rothstein argues that residential segregation in American cities was produced by government policy, not private preference or market outcomes. If that's true, what obligation does an urban product company have with respect to the spatial patterns that history created?

Normative question without a clean answer. The goal is for students to articulate a position and defend it, not to produce consensus.

- 2** A product can fail on equity in at least three ways: by design (it can't be used by low-income people), by deployment (it's only deployed where rich people live), or by displacement (it raises housing costs). Which is easiest to fix? Which is hardest? Why?

Displacement is hardest because it is a systems-level effect. Design is easiest. Deployment is in between — it's a business model question.

- 3** Klinenberg argues that social infrastructure — libraries, parks, community centers — is the key to urban resilience and equity. Can a private product substitute for social infrastructure? When does it complement it? When does it crowd it out?

The privatization of public goods question. Particularly relevant to sharing economy platforms.

GRADUATE / MASTER'S QUESTIONS

- G1** Eubanks documents how algorithmic systems consistently harm the poorest communities, often without explicit discriminatory intent — through proxy variables, missing data, and misaligned incentives. How would you audit an urban product for this kind of harm? What would a meaningful audit look like?

- G2** Grengs argues that conventional transportation planning metrics — vehicle miles traveled, average travel time — systematically misrepresent the accessibility experience of low-income and minority travelers. What metrics would better capture that experience? What data would you need to calculate them?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students sometimes read the equity discussion as a constraint on product design. Frame it instead as a design specification: equity is a feature requirement, not a filter. Products that meet it are more durable and more defensible than those that don't.

The 'lift all boats' argument — that growth and innovation are good for everyone in the long run — is a common response to the displacement critique. Engage it seriously: the empirical evidence on whether urban tech products lift boats for low-income residents is mixed at best.

Graduate students may apply equity frameworks categorically — every product is extractive. The productive question is: under what conditions can a private product genuinely advance equity?

CONNECTIONS TO OTHER SESSIONS

Equity is not a standalone session — it is a lens applied in every subsequent session. In Week 9 (regulation), equity considerations shape what cities require of products as a condition of operation. In Week 11 (climate resilience), climate impacts are inequitably distributed. In Week 12 (pitch workshop), teams must specify their equity commitment. In Week 15 (final pitch), equity is an evaluation criterion.

FACILITATOR NOTES

The field session data is the best opening for this session if it was collected the week before. Students often discover equity gaps in their own observation data that they did not consciously notice at the time — returning to that data with an equity lens is a powerful pedagogical moment.

Week 9 · Regulation as Design

FORMAT

Seminar

TIMING

75 min

LEVEL

UG + Grad

SESSION OVERVIEW

For product teams, regulatory requirements often feel like external constraints — arbitrary rules that slow deployment and add cost. This session inverts that frame: regulation is itself a form of design, encoding collective decisions about how a city should work. Understanding regulation as design makes it legible, navigable, and sometimes improvable. The goal is for students to leave as sophisticated readers of regulatory text who can engage municipal partners as peer designers, not adversaries.

THEORETICAL FRAME

The Sidewalk Toronto case (O'Kane) is the most instructive recent example of a product company that treated regulation as an obstacle rather than a design partner — and failed accordingly. Green's 'Smart Enough City' provides the principled alternative: technology should serve democratic governance, not circumvent it. Lydon's tactical urbanism shows that regulatory creativity can be generative, not just compliant.

SUGGESTED DISCUSSION ARC

0–10 min

Show the actual text of a scooter permit requirement from a major U.S. city. Ask: 'What problem is each requirement solving? Who advocated for it? Who opposed it? What product decisions does it constrain?' Work through it clause by clause.

10–20 min

Brief lecture: three modes of regulatory engagement — adversarial (operate and ask forgiveness — Uber's early strategy), collaborative (work with cities to design the framework — SoBi's strategy), and creative compliance (design exactly to what the regulation requires, often the most innovative approach).

20–50 min

Seminar on Bertaud, Green, and Lydon. Questions 1–3. Question 2 (creative compliance) is most generative for product designers.

50–65 min

Graduate section: O'Kane (Sidewalk Toronto), Frug, Goldsmith. Questions G1 and G2.

65–75 min

Regulatory simulation: each team receives a fictional municipal RFP for a mobility product. Teams have 15 minutes to identify (1) the city's actual goals behind the stated requirements, (2) one way to exceed the requirements, and (3) one way the requirements conflict with each other.

DISCUSSION QUESTIONS — ALL STUDENTS

- 1** Bertaud argues that land use regulations often aim to redesign cities rather than alleviate clear negative externalities — and that when they do, they produce bad outcomes. Apply that argument to a regulation on your proposed urban product. Is it alleviating an externality or trying to redesign the city?

Most regulations do both simultaneously. The productive question is: which function dominates, and does the regulation achieve it?

- 2** Lydon's 'tactical urbanism' describes small-scale, temporary interventions that operate in regulatory gaps — a pop-up bike lane, a parklet, a weekend street closure. What can tactical urbanism do that a formal product launch cannot? What are its limits?

Tactical urbanism as prototyping: it creates evidence that changes what is politically possible. But it requires either regulatory tolerance or calculated rule-bending. Both have limits.

- 3** Uber's early growth strategy was 'operate and ask forgiveness' — launch without regulatory approval, build user base, then use political pressure to force legalization. Does that strategy work? What does it cost? Who pays?

It worked for Uber in ride-hailing. It did not work for Bird or Lime in many cities. The strategy is more available to well-funded platforms than to small operators. Ask what the strategy implies about the relationship between capital and regulation.

GRADUATE / MASTER'S QUESTIONS

- G1** O'Kane's account of Sidewalk Toronto shows a product company that misread municipal governance — treating elected officials and public agencies as obstacles to be managed rather than partners in democratic decision-making. What specifically did Sidewalk Labs get wrong, and how would a different regulatory strategy have changed the outcome?

- G2** Frug argues that American cities are legally weak — constrained by state law in ways that European cities are not — and that this weakness explains why cities often fail to regulate effectively. Does that argument change how you think about regulatory strategy for an urban product? Does it make you more or less sympathetic to cities as product partners?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

The 'move fast and break things' regulatory strategy is celebrated in tech culture. Students should understand its costs: confrontational approaches produced bans, not permission, in many cities. Those most receptive to micromobility innovation are cities where operators engaged collaboratively.

Regulatory compliance is not the ceiling — it is the floor. The most successful urban products often exceed regulatory minimums in ways that build trust and create competitive advantage. Regulation is a design brief, not a checklist.

Graduate students may treat regulatory analysis as purely normative. The productive question is empirical: does this regulation achieve its stated purpose? Often the answer requires evidence that regulators don't have.

CONNECTIONS TO OTHER SESSIONS

Regulation returns in Week 13 (go-to-market — regulatory strategy is part of the city-by-city expansion plan) and Week 14 (failure cases — Sidewalk Toronto's failure was fundamentally regulatory). The Week 9 regulatory simulation is the dress rehearsal for the Week 12 pitch workshop, where teams must specify their regulatory strategy.

FACILITATOR NOTES

The regulatory text exercise works best with a real document — any major American city's scooter permit or bikeshare concession agreement is publicly available. San Francisco, Los Angeles, and Chicago all have instructive examples. The act of reading actual regulatory language — which is often surprisingly specific and sometimes self-contradictory — is itself educational.

Week 10 · Civic Tech

FORMAT
Seminar

TIMING
75 min

LEVEL
UG + Grad

SESSION OVERVIEW

Civic technology — software that supports democratic participation, government service delivery, or civic engagement — represents the most direct intersection of product design and urban governance. This session examines what civic tech can and cannot do: it can make government services more accessible, create new channels for citizen input, and generate data about public needs. It cannot substitute for political will, repair defunded public services, or overcome structural inequality through better interfaces.

THEORETICAL FRAME

Green's critique of civic tech solutionism — the belief that better apps can solve political problems — is the session's intellectual edge. Morozov's broader critique of solutionism provides the theoretical frame. Fung's work on democratic participation offers the corrective: civic tech works best when it supports genuine deliberation and shared decision-making, not just data collection.

SUGGESTED DISCUSSION ARC

0–10 min

Ask: 'Has anyone used a government app or website in the last month? What for? Did it work?' Collect responses. Map the gap between what digital government promises and what it delivers.

10–20 min

Brief lecture: three generations of civic tech — (1) e-government (moving paper forms online), (2) open data (publishing government data for civic hackers), (3) participatory platforms (311, participatory budgeting, planning platforms). Each generation has delivered differently from its promise.

20–50 min

Seminar on Green, Townsend, and Lerner. Questions 1–3. Question 2 (what civic tech cannot do) is the most important.

50–65 min

Graduate section: Schrock, Fung, Morozov. Questions G1 and G2. The Morozov question is deliberately provocative — let it run.

65–75 min

Design exercise: design the civic engagement feature for your Week 12 pitch product. What decisions will community members make? What information will they need? What is the minimum viable version of genuine participation?

DISCUSSION QUESTIONS — ALL STUDENTS

- 1** **311 systems let residents report potholes, broken streetlights, and other service failures, producing enormous data about where infrastructure is failing. Does that data make cities more equitable, or does it advantage residents with smartphones and time to report?**

311 data consistently shows wealthier, whiter neighborhoods generate more reports — partly because they have more time and digital access, partly because they have a stronger expectation of government responsiveness. Is that a problem with 311, or with the city's response function?

- 2** **Morozov argues that solutionism — the belief that every social problem has a technological solution — is the signature mistake of civic tech. What are the things a great civic tech product genuinely cannot do? Name three.**

Students typically produce: cannot generate political will, cannot repair defunded services, cannot overcome structural racism. Push them to be specific.

- 3** **Porto Alegre's participatory budgeting system — letting residents vote on how to allocate a portion of the city's capital budget — is the most successful example of civic tech enabling genuine democratic participation. What made it work? What limits its generalizability?**

Key factors: strong civil society organizations, a sympathetic government, started small before scaling. The digital version in many cities has been less successful. Ask why.

GRADUATE / MASTER'S QUESTIONS

- G1** Schrock argues that civic tech has a class and race problem — primarily built by and for educated, digitally fluent, relatively privileged users. What does that demographic profile imply about whose problems get solved? How would you design differently?

- G2** Fung's work on 'empowered participation' describes conditions under which citizen engagement actually influences decisions. Apply his framework to a civic tech product you know. Does it meet his conditions? If not, what would it take?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often confuse engagement (giving citizens a channel to speak) with participation (giving citizens genuine decision-making power). Most civic tech produces the former while calling it the latter. The distinction matters enormously.

Participatory budgeting is sometimes presented as a panacea. Its successes are real and significant; so are its limits (typically covers 1–5% of municipal capital budgets, not operating budgets). Keep the enthusiasm grounded in scale.

Graduate students sometimes treat Morozov's solutionism critique as a reason not to build civic tech. It

is an argument for building civic tech more carefully and honestly — with clear-eyed claims about what it can and cannot do.

CONNECTIONS TO OTHER SESSIONS

Civic tech is the public sector version of the product design challenge the course has been building toward. Connect to Week 3 (data — civic tech generates data; whose data is it?), Week 8 (equity — who uses civic tech?), Week 9 (regulation — civic tech must navigate government procurement and approval), and Week 12 (pitch — teams should consider whether their product has a civic partnership component).

FACILITATOR NOTES

The civic engagement design exercise works best if teams have their Week 12 pitch concepts in rough form. If they don't, use a fictional product brief. The minimum viable participation question — 'what is the least amount of participation that is still genuine?' — often produces the most useful product design insight.

Week 11 · Climate & Resilience**FORMAT**
Seminar**TIMING**
75 min**LEVEL**
UG + Grad**SESSION OVERVIEW**

Cities are simultaneously major contributors to climate change and the primary victims of its effects. Urban products that address climate — through emissions reduction, energy efficiency, or resilience infrastructure — operate in a context where the stakes are existential and the policy environment is rapidly changing. This session asks what products can contribute to urban climate response, and what only policy can achieve.

THEORETICAL FRAME

The session resists both techno-optimism (better apps will solve climate change) and techno-pessimism (nothing short of systemic transformation matters). Chakrabarti's density argument is the most product-relevant: compact, walkable, transit-served development is the most effective single intervention for urban emissions, and products that enable density are climate products. Klinenberg's social infrastructure argument adds the resilience dimension.

SUGGESTED DISCUSSION ARC**0–10 min**

Show the IPCC urban emissions chart: cities account for ~70% of global CO₂ emissions. Ask: 'Where in that 70% does a product intervention land? What's the maximum contribution of the best conceivable urban product?' Surface: most product interventions are marginal relative to the challenge — and that's an honest assessment, not a counsel of despair.

10–20 min

Brief lecture: four categories of urban climate product — (1) emissions reduction (transit, EV infrastructure, energy efficiency), (2) carbon sequestration (urban forestry, green roofs), (3) resilience infrastructure (flood barriers, cooling centers, early warning systems), (4) adaptive governance (tools that help cities make faster climate decisions).

20–50 min

Seminar on Klinenberg, Chakrabarti, and Mostafavi. Questions 1–3. Question 2 (density as climate policy) is the most politically charged.

50–65 min

Graduate section: Bulkeley, Vale, Beatley. Questions G1 and G2.

65–75 min

Climate product audit: teams apply a simple framework to their pitch product — (1) what are the product's direct emissions? (2) what emissions does it reduce? (3) does it make the city more or less resilient to climate impacts? Honest answers, not marketing.

DISCUSSION QUESTIONS — ALL STUDENTS

- 1 **Klinenberg argues that social infrastructure is the key to urban resilience. During the 1995 Chicago heat wave, neighborhoods with the lowest death rates were those with the densest social infrastructure, not those with the best air conditioning. What does that finding imply for product design in a climate-stressed city?**
The implication is counterintuitive: resilience is more social than technical. Products that strengthen social infrastructure may do more for climate resilience than cooling technology.
- 2 **Chakrabarti argues that density is the single most powerful climate intervention available to cities — more people per square mile means less energy per person. But density is politically unpopular in many cities. Can a product make density more politically acceptable? Should it try?**
This bridges product design and political strategy. What are the ethics of a product company taking a position on land use?
- 3 **Hurricane Katrina demonstrated that urban resilience is not evenly distributed — wealthy neighborhoods recovered faster and more completely than poor ones. Does the same pattern hold for other climate impacts? What product or policy intervention would change that pattern?**
The answer is yes, consistently. The more interesting question is mechanism: is it access to resources, social networks, housing quality, or political power that determines differential recovery?

GRADUATE / MASTER'S QUESTIONS

- G1 **Vale and Campanella's 'Resilient City' shows that cities always recover from disasters — the question is who recovers, how fast, and to what. Does that finding make you more or less optimistic about urban climate resilience? Does it change what you think a product should do?**
- G2 **Bulkeley argues that urban sustainability transitions require alignment across government, markets, civil society, and residents — no single actor can drive them alone. Where does a private product fit in that multi-actor framework? What can it do that the other actors cannot?**

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

The '70% of emissions from cities' statistic is sometimes used to argue that urban products can solve climate change. The honest argument is that urban policy — density, transit, electrification — is a major lever, and products can support or undermine it. That is important, but it is not the same as solving climate change.

Resilience is sometimes conflated with robustness. Resilience properly understood is the ability to absorb shocks and reorganize — it implies change, not preservation. Products designed for robustness often fail to support genuine resilience.

Graduate students may treat the equity and climate agendas as identical. They are related but not the same: the most emissions-efficient policies can be implemented equitably or inequitably. The intersection requires explicit design attention.

CONNECTIONS TO OTHER SESSIONS

Climate and resilience connects to Week 8 (equity — climate impacts are inequitably distributed), Week 9 (regulation — climate policy is the fastest-changing regulatory environment urban products face), and Week 12 (pitch — climate impact is an increasingly expected component of any urban product pitch).

FACILITATOR NOTES

The climate product audit is most effective when students are honest about negative impacts as well as positive ones. Model this: 'A dockless scooter reduces car trips (good) but requires charging trucks driving around the city (not so good) and has a short product lifetime before becoming waste (bad). Net assessment?' The habit of honest accounting is as important as the specific answers.

Week 12 · Product Pitch Workshop

FORMAT

Workshop + critique

TIMING

90 min

LEVEL

UG + Grad

SESSION OVERVIEW

The pitch workshop is the applied center of the course's second half. Teams present early urban product concepts — not polished pitches but honest rough cuts — for structured critique from peers and invited practitioners. The instructor's goal is to push: to identify unexamined assumptions, unspecified equity commitments, unthought regulatory strategies, and unasked business model questions.

THEORETICAL FRAME

The pitch format forces students to synthesize the course's frameworks into a coherent product proposition. Strong pitches will have a clear city-as-system insight (Week 1), an equity commitment (Week 8), a regulatory strategy (Week 9), and a climate assessment (Week 11). Pitches that lack any of these will be visibly incomplete — which is the point of this session.

SUGGESTED DISCUSSION ARC

0–10 min

Brief framing: 'A pitch is not a presentation — it is a conversation. The goal is not to persuade but to learn what questions you haven't answered yet. The most valuable thing a panel can do is find your unexamined assumptions.'

10–70 min

Team pitches: 8 minutes per team + 12 minutes structured critique. Critique protocol: (1) What is the product? (2) Who is the user? (3) Who is the city partner? (4) What does it do to the system? (5) What is the equity commitment? (6) What is the regulatory strategy? (7) What would falsify your demand assumption?

70–85 min

Plenary synthesis: 'Across these pitches, what equity commitment was most specific? Least? What regulatory strategy seemed most realistic? What assumption appeared in every pitch that we should challenge together?'

85–90 min

Next steps: final pitch is Week 15. Teams have three weeks. Each team identifies their most urgent gap to develop.

DISCUSSION QUESTIONS — ALL STUDENTS

1

What specific city are you launching in first, and why that city? What does that city's regulatory environment, transit infrastructure, and demographic composition mean for your product?

Generic answers indicate insufficient market research. Push for specificity: San Francisco's scooter

regulations, Chicago's transit coverage gaps, Atlanta's equity challenges are specific facts that should shape the pitch.

2 **Who is your city partner — the agency, office, or official whose support you need? What do they want, and how does your product serve their goals, not just your users' goals?**

Urban products require institutional partners, not just users. Many student pitches lack this dimension entirely. A product with no institutional home in city government has no path to scale.

3 **What does your product do to the urban system? What feedback loops does it create? What unintended consequences have you considered?**

Systems question from Week 1. Teams that have absorbed the course's framing will have thought about this. Those that haven't will produce blank stares — which is valuable diagnostic information.

GRADUATE / MASTER'S QUESTIONS

G1 What is the minimum viable version of your equity commitment — the thing you would do even if you had no additional resources? What is the maximum viable version — what you would do if equity were your primary design constraint?

G2 Describe your product's business model in the format of a platform economics analysis: who are the two sides of the market, what is the data extraction mechanism, what are the network effects, and what are the externalities you are not currently pricing?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Teams often pitch the ideal version of their product rather than the version they can actually build. Push for honesty about constraints: 'In year one, with \$2M and a team of 8, what can you actually deliver?'

The equity commitment is often vague. Push for specificity: which communities, what commitments, measured how, enforced by whom?

Graduate teams sometimes over-theorize and under-specify. Push for concrete user journeys, specific city partners, and actual regulatory requirements.

CONNECTIONS TO OTHER SESSIONS

The pitch workshop synthesizes Weeks 1–11. The final pitch (Week 15) is the fully developed version of what is presented here. Teams should leave with a clear prioritized list of what to develop in Weeks 13 and 14.

FACILITATOR NOTES

Invite 2–3 practitioners as pitch critics: a city planner, a mobility product manager, and a civic tech practitioner make an ideal panel. Brief them with the evaluation criteria in advance. The practitioner perspective is most valuable when it is specific and honest.

Week 13 · Go-to-Market in Cities

FORMAT
Seminar

TIMING
75 min

LEVEL
UG + Grad

SESSION OVERVIEW

City-by-city expansion is fundamentally different from the standard tech go-to-market playbook. Each city has its own regulatory environment, political culture, transit infrastructure, demographic composition, and civic partner landscape. Strategies that work in San Francisco may fail in Atlanta; what is legal in Denver may be prohibited in New York. This session gives students the analytical tools to build a city-specific expansion strategy — not as a growth hacking exercise, but as a genuine engagement with urban complexity.

THEORETICAL FRAME

Rogers's diffusion of innovations framework applies to cities as innovation systems: early adopter cities lower the barrier for later adopters by creating regulatory precedent and market evidence. Hoffman's @arcktip go-to-market framework grounds this in the specific experience of building Social Bicycles across multiple cities simultaneously.

SUGGESTED DISCUSSION ARC

0–10 min

Ask: 'If you were launching your pitch product tomorrow, which city would you choose and why? What's your second choice?' Collect answers. Surface the range of criteria students are applying — then ask them to make those criteria explicit.

10–20 min

Brief lecture: five dimensions of city selection — regulatory openness, infrastructure fit, demographics, political alignment, and competitive landscape. Walk through how each applies to a specific product (dockless bikes works well).

20–50 min

Seminar on Hoffman, Rogers, and Moore. Questions 1–3. Question 2 (crossing the chasm in cities) is the most product-relevant.

50–65 min

Graduate section: Mazzucato, Sassen, Goldsmith. Questions G1 and G2.

65–75 min

Teams build a city selection scorecard for their pitch product: five criteria, weighted by importance, applied to three candidate cities. Teams share their top choice and the criteria that drove it.

DISCUSSION QUESTIONS — ALL STUDENTS

1

Rogers describes early adopters as willing to accept risk in exchange for being first. Who are the early adopter cities for urban tech products? What makes a city an early

adopter? And what does it take to move beyond them?

Portland, San Francisco, Denver, Amsterdam, and Singapore recur as early adopter cities across mobility, civic tech, and smart city categories. Ask: what do they have in common? (Strong cycling culture, progressive governance, density, high income.) What does that profile leave out?

2 Moore's 'chasm' describes the gap between early adopters and the pragmatist majority. For urban products, that chasm is both a market challenge and a governance challenge — the cities that most need your product are often the least able to pilot it. How do you cross that chasm?

Honest answers often involve: public subsidy (foundation or government grants for equity pilots), regulatory change (working with standards bodies to lower the bar for mainstream adoption), or product simplification (making the product work in lower-capacity environments).

3 Describe the ideal city partner relationship for your pitch product. Who in city government do you need to work with? What do they get from the partnership that they can't get elsewhere? What do you get?

The partnership must be genuinely bilateral — the city gets something it values, not just a product that exists in its jurisdiction. The best partnerships are ones where the city's problem is the product's value proposition.

GRADUATE / MASTER'S QUESTIONS

G1 Mazzucato argues that the state is the most important risk-taking actor in the innovation ecosystem — markets only absorb risk after governments have de-risked technologies. Does her argument apply to urban tech? What has government investment made possible that markets couldn't have produced alone?

G2 Sassen's 'Global City' describes a hierarchy of cities in which a small number concentrate the command functions of the global economy. Does that hierarchy shape which cities are receptive to urban tech products? What does it mean to build for cities outside that hierarchy?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students often choose their launch city based on where they would like to live. Push them to choose based on product-market fit: which city has the problem your product solves, the regulatory environment that allows it, and the partners that can help it scale?

The 'move to the next city' playbook assumes cities are substitutable markets. They are not. Regulatory approval in Denver does not transfer to Atlanta. This is not a bug — it is the nature of urban governance — but it has real implications for growth strategy.

Graduate students sometimes treat go-to-market as purely strategic and ignore equity implications. Who is helped and who is not by a city-selection strategy that prioritizes regulatory ease and high

income? This is an ethics question, not just a strategy question.

CONNECTIONS TO OTHER SESSIONS

The go-to-market session is the bridge between the analytical content of Weeks 1–11 and the final pitch in Week 15. Teams should leave with a specific city selection, a specific city partner, and a regulatory strategy.

FACILITATOR NOTES

The city selection scorecard exercise is most effective when teams apply it honestly — including to cities where their product would face the most resistance. The tension between 'where I could launch easily' and 'where the need is greatest' is exactly the equity tension the course has been building toward. Surface it explicitly.

Week 14 · Failure Cases

FORMAT

Case study seminar

TIMING

75 min

LEVEL

UG + Grad

SESSION OVERVIEW

Every field learns more from its failures than its successes. This session examines three high-profile urban product failures — Sidewalk Toronto (Alphabet's smart neighborhood project), WeWork (co-working as urban product at scale), and Bird's rapid collapse in many cities — to extract generalizable lessons about why urban products fail and what failure teaches us about cities.

THEORETICAL FRAME

Scott's 'Seeing Like a State' provides the theoretical frame for all three cases: each failure involved a high-modernist impulse to simplify and optimize a complex urban system from above. Sidewalk Toronto tried to redesign urban governance. WeWork tried to redesign the relationship between work and place. Bird tried to override municipal regulatory authority. All three underestimated the complexity of the systems they were entering.

SUGGESTED DISCUSSION ARC

0–10 min

Before revealing the case studies: ask teams 'What do you think is the most likely reason your pitch product will fail?' Collect answers honestly. Then: 'Today's cases are a stress test of those predictions.'

10–30 min

Sidewalk Toronto case: brief summary (O'Kane), then Questions 1 and G1. The regulatory and governance failure is the main lesson.

30–50 min

WeWork and Bird: brief summaries, then Questions 2 and 3. Business model and over-extension lessons.

50–65 min

Graduate section: Flyvbjerg, Scott (Ch. 4), Townsend (Ch. 9). Question G2.

65–75 min

Close: teams revise their 'most likely reason to fail' answer in light of today's discussion. What changed? What is your product's Sidewalk Toronto risk? Its WeWork risk? Its Bird risk?

DISCUSSION QUESTIONS — ALL STUDENTS

1

Sidewalk Labs' Toronto project was cancelled because of a combination of privacy concerns, governance disputes, and financial uncertainty during COVID. Which of those three do you think was the most fundamental cause? What would success have

required?

The governance dispute was deepest: Sidewalk Labs wanted to collect and own urban data generated in public space. Toronto residents and officials refused. This is a first-principles disagreement about who owns the city, not a solvable technical or financial problem.

- 2 WeWork's collapse revealed that its product — flexible co-working space in cities — had genuine value but a fundamentally broken business model. How do you distinguish between a product that fails because the product is wrong and one that fails because the business model is wrong?**

The diagnostic question: are customers using it? (Yes, WeWork had high occupancy.) Are customers paying enough to cover costs? (No.) The product was real; the economics were broken.

- 3 Bird deployed scooters in dozens of cities without regulatory approval, built user bases, and then sought permits. In many cities the strategy failed — the cities banned or severely restricted the product. What was Bird's error? What should they have done instead?**

Bird read the Uber playbook and applied it to a product requiring physical presence in public space. Unlike Uber (which ran on private roads and parking), Bird occupied sidewalks — which are unambiguously public. The regulatory calculus was different, and Bird didn't adjust.

GRADUATE / MASTER'S QUESTIONS

- G1** O'Kane's account of Sidewalk Toronto reveals a fundamental tension: Sidewalk Labs wanted to innovate beyond existing regulatory frameworks; Toronto's public agencies wanted to maintain democratic accountability over public space. Is that tension resolvable? Or does it mean that certain kinds of urban tech are simply incompatible with democratic governance?
- G2** Flyvbjerg finds that large-scale urban projects consistently underestimate costs and overestimate benefits — not by accident but because of systematic optimism bias and strategic misrepresentation. Does that finding apply to urban product startups? What would honest forecasting look like?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Students sometimes read the failure cases as evidence that urban tech is impossible. The cases show something more specific: failures of regulatory strategy, governance design, and business model discipline. Each failure was avoidable — not inevitable.

Sidewalk Toronto is sometimes read as a story about privacy. The deeper story is about who governs public space. The privacy concerns were real, but they were the visible symptom of a deeper governance dispute.

WeWork and Bird failed for different reasons. Conflating them under 'urban tech hype' misses the specificity: WeWork's problem was capital structure and accounting; Bird's was regulatory strategy.

CONNECTIONS TO OTHER SESSIONS

The failure cases directly inform final pitches: teams should be able to explain how their product avoids the Sidewalk Toronto governance trap, the WeWork business model trap, and the Bird regulatory trap. In Week 15, panels will ask these questions explicitly.

FACILITATOR NOTES

The opening question — 'what is your product's most likely failure mode?' — works best if collected anonymously before the case studies are revealed, then returned after. The contrast between students' pre-case predictions and their post-case revisions is itself instructive. Many students will discover they predicted a technical failure and should have predicted a governance or business model failure.

Week 15 · Final Pitch Day

FORMAT

Public pitch + external panel

TIMING

120 min

LEVEL

UG + Grad

SESSION OVERVIEW

Final pitch day is a public event. Teams present fully developed urban product concepts to a panel of city planners, urban practitioners, and investors. The evaluation criteria reflect the course's full analytical framework: urban systems insight, equity commitment, regulatory strategy, city partner relationship, business model, and climate assessment.

THEORETICAL FRAME

No new theory. The session tests whether students can synthesize the course's full analytical toolkit — complex systems, planning tradition, data critique, mobility economics, platform logic, equity frameworks, regulatory strategy, civic partnership, and climate accountability — into a coherent and honest product proposition.

SUGGESTED DISCUSSION ARC

0–5 min

Frame: 'This is not a competition — it is a demonstration of what you now know about building products in cities. The best pitch is the most honest one: clear about what the product can do, rigorous about what it cannot, and specific about the city it serves.'

5–90 min

Presentations with panel critique. For a class of 15: 5 UG teams at 20 min each = 100 min; adjust for class size. Reserve 20 minutes for synthesis.

90–110 min

Panel synthesis: each panel member offers one observation about the cohort as a whole — what analytical move appeared in every strong pitch? What gap appeared in every weak one?

110–120 min

Instructor close: return to Week 1's question — 'What does a product do to a city?' — and answer it with specific evidence from the pitches presented today.

DISCUSSION QUESTIONS — ALL STUDENTS

1

Which of the course's frameworks — complex systems, platform logic, equity analysis, regulatory strategy — is doing the most work in this pitch? Where is that framework's limitation showing up?

Strong pitches will have used multiple frameworks in tension. Weak pitches will have used one framework as a hammer and treated every problem as a nail.

2 | If this product scales successfully across 20 cities over 5 years, what does the city look like? What has changed? What has stayed the same? What has gotten worse?

Systems consequence question. The best pitches will have thought about this honestly, including the negative consequences.

3 | What would make you abandon this product? Under what conditions would you conclude that it was doing more harm than good, and what would you do then?

The exit ethics question. Urban products operate in public space with public consequences. What accountability mechanism exists for the people who built it?

GRADUATE / MASTER'S QUESTIONS

G1 | Your product has a city partner. In five years, if the product succeeds, who has more power: the city or the product company? Is that the right answer?

G2 | Looking back at the course: which reading most changed how you think about cities, and why? How does that change show up in your pitch?

COMMON MISCONCEPTIONS & HOW TO ADDRESS THEM

Teams sometimes interpret 'fully developed' as 'polished.' A polished pitch with unexamined assumptions is less valuable than a rough pitch with honest uncertainty acknowledged. Reward intellectual honesty.

Panels sometimes default to investor-mode critique — asking about market size, valuation, competition. Redirect to the course's framework: equity, governance, system effects.

CONNECTIONS TO OTHER SESSIONS

The final pitch is the answer to the question asked in Week 1: 'What does a product do to a city?' The course has given students 15 weeks of analytical tools to answer that question honestly and rigorously. The pitch is the proof.

FACILITATOR NOTES

Panel composition: a city planner (ideally from the course city's planning department or transportation agency), a mobility or civic tech practitioner, and optionally a mission-aligned investor or foundation program officer. Brief panel members with evaluation criteria 48 hours in advance. If Patrick Hoffman is available as a panel member, his practitioner perspective — especially on regulatory strategy and go-to-market — is uniquely valuable. The best final session feels like the beginning of a real project, not the

end of a course.

APPENDIX: QUICK-REFERENCE QUESTION BANK BY THEME

These questions can redirect or deepen discussion in any session:

Urban Systems (any session)

- What does this product do to the urban system? What feedback loops does it create?
- What would Jacobs say about this product? What would Bertaud say?
- What does optimizing for the measurable sacrifice that is not measurable?

Equity & Justice (Weeks 5, 8, 11, 13)

- Who benefits from this product in year one? In year five? Are they the same people?
- Which equity failure mode applies here: design, deployment, or displacement?
- What would Sheller's mobility justice framework require of this product as a condition of legitimacy?

Regulation & Governance (Weeks 2, 9, 14)

- Is this regulation alleviating an externality or redesigning the city? What is the difference in this case?
- What does the regulatory text reveal about what the city actually values?
- What would a collaborative — rather than adversarial — regulatory strategy look like?

Data & Measurement (Weeks 3, 10)

- What does this product measure, and what does optimizing for that metric make worse?
- Who generated this data, who owns it, and who can see it?
- What would Loukissas say about applying this dataset in a different city context?

Business Model & Capital (Weeks 6, 12, 14)

- Who internalizes the profits and who absorbs the externalized costs?
- What happens to this product's mission if it is acquired by a larger platform?
- What is the minimum business model that could sustain this product without extracting value from the city?

Practitioner Perspective (Weeks 5, 13, 15)

- What does Patrick's account of building JUMP see that academic readings cannot? What does it miss?
- What would you do differently from the SoBi/JUMP arc, given what you now know?
- If you were advising a city council on whether to permit this product, what three questions would you ask the product team?

This guide is a living document. Faculty who use it are encouraged to annotate, revise, and return observations to Patrick Hoffman (@arcktip) so that future iterations benefit from classroom experience across institutions.